

Team default - Design Rationale

2026 Protein Design in SynBio Challenges

1. Team Strategy

Surface charge engineering + Progressive stabilization + Chimera design

- TGP surface: K79E, Q204E add Glu negative charges (Close 2015)
- Progressive FoldX: 3 mutations (min) vs 7 mutations (mod) on amacGFP
- Brightness combo: K167M (4.57, brightest mutant in dataset)
- Cross-species chimera: sfGFP+amacGFP+TGP+usGFP hybrid

2. Pipeline

Identical 6-stage pipeline as Team 32-603 (shared codebase).

- Stages 1-5: Data, FoldX, Literature, Design, Validation - same framework
- Stage 6: Team split with different design philosophy applied

3. Agent Decision Points (Divergent from Team 32-603)

Decision 1: Surface Engineering Priority

IF scaffold is already ultra-stable (ppluGFP DG=-49.88):

- > Prioritize surface charge over stability mutations
- > Add Glu at 3 surface positions: S73E, S147E, S174E
- > Rationale: anti-aggregation under 72C thermal stress [Close 2015]

Decision 2: Progressive Stabilization on amacGFP

FOR amacGFP scaffold (WT DG=+3.55, brightness=3.97):

- > Variant A (min): 3 high-confidence mutations (Y39N, N149K, E172K)
- > Variant B (mod): 7 mutations (add usGFP core + extra FoldX)
- > Rationale: compare minimal vs moderate perturbation on same scaffold

Decision 3: Cross-Species Chimera

Instead of single-scaffold engineering:

- > sfGFP backbone: competition benchmark, proven folding
- > amacGFP chromophore: 1.8x avGFP intrinsic brightness
- > TGP surface: K79E, Q204E for anti-aggregation
- > usGFP core: Q69L, N164Y for hydrophobic packing
- > Rationale: combine best features from 4 GFP families

Decision 4: Brightness Combo on cgreGFP

cgreGFP = brightest WT (4.50) AND stable (DG=-26.89)

- > Pure brightness: K167M(4.57) + D170E(4.56) + K8Q + L197P
- > No stability mutations needed, only brightness boost

4. Design Details

1. sfGFP_Plus_v2_TGP

sfGFP (2B3P) | Len: 238 | WT DG: +3.40

Muts: Q69L, K79E(TGP), N149K, N164Y, E172K, Q204E(TGP)

TGP surface Glu at K79E+Q204E. Anti-aggregation + core stability.

2. amacGFP_v1_min

amacGFP (7LG4) | Len: 238 | WT DG: +3.55

Muts: Y39N(-4.36), N149K(-4.06), E172K(-3.06)

Minimal 3-mutant approach. Y39N alone turns DG +3.55 to -0.81.

3. amacGFP_v2_mod

amacGFP (7LG4) | Len: 238 | WT DG: +3.55

Muts: Y39N, Q69L(usGFP), S147P, N149K, N164Y(usGFP), E172K, D190N

Moderate 7-mutant approach. Compares with Team 32-603 v3_max.

4. ppluGFP_v2_surface

ppluGFP (2G6X) | Len: 222 | WT DG: -49.88

Muts: S73E, S147E, S174E (3x surface Glu), L206E, Y88F, S148A, R211F

Ultra-stable scaffold + surface Glu engineering.

5. cgreGFP_v2_bright

cgreGFP (2HPW) | Len: 235 | WT DG: -26.89

Muts: K8Q, K167M(4.57), D170E(4.56), L197P

Pure brightness combo. K167M = best single brightness mutant.

6. Chimera_sf_amac

sfGFP+amacGFP | Len: 238 | WT DG: hybrid

Muts: sfGFP scaffold + amacGFP chromophore + TGP surface + usGFP core

Most innovative: combines 4 GFP families into one sequence.

5. Summary

1	sfGFP_TGP	sfGFP	238	Surface eng
2	amacGFP_v1	amacGFP	238	Min FoldX(3)
3	amacGFP_v2	amacGFP	238	Mod FoldX(7)
4	ppluGFP_surf	ppluGFP	222	Surface eng
5	cgreGFP_v2	cgreGFP	235	Brightness
6	Chimera	sf+amac	238	Chimera

Seq

6. References

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3. Scott et al. 2018. Sci Rep 8:159
4. Sarkisyan et al. 2016. Nature 533:397-401
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6. Hirano et al. 2022. Nat Biotechnol 40:1132-1142
7. Zhang et al. 2024. Nat Methods 21:657-665
8. GitHub: <https://github.com/eenneenn-crypto/GFP-Design-SynBio2026>